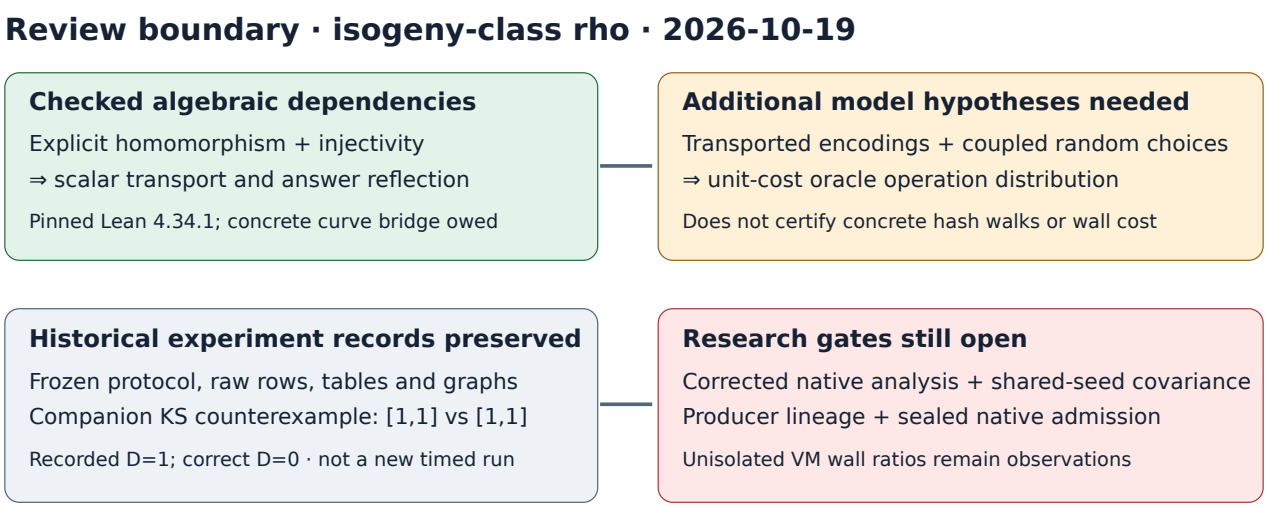
Isogeny-class rho: review correction

2026-10-19 · PR 2187 · historical run-20261018T0857Z

This is a source and claim audit. No new timed experiment or corrected historical statistical result is claimed. The frozen protocol and raw rows remain unchanged.



Sources: REVIEW-20261019.md · PROOF-MAP.md · companion source a277c5f · historical run-20261018T0857Z

What changed

The manuscript and README now enumerate the ancillary flags: one B1 generator, two P1 shift cells, one P0 shift cell, and B1 generator Kruskal–Wallis p=0.039. Primary comparisons inside their reported bands do not justify “no test fired.” The historical 1.7–2.2× wall-time ratio is an unisolated shared-VM observation, not an admitted performance result.

The native KS counterexample binds companion source a277c5f: identical samples [1,1] and [1,1] produce D=1 instead of 0. Rank-test ties and shared-seed covariance also require review. Historical p-values and detector resolution remain provisional until corrected native analysis is bound to the actual producer source and frozen rows. No preregistered test was silently replaced.

Proof and model scope

The pinned Lean project checks scalar transport and answer reflection under explicit homomorphism and injectivity hypotheses, and proves the mod-8 number-theoretic implication from explicit discriminant/divisibility hypotheses using pinned Mathlib. PROOF-MAP.md states the checked scope and remaining bridges. Concrete isogeny/group-law/native-code refinement, coupled oracle costs, characteristic-two map certification and experiment admission are separate obligations. The mod-8 restriction alone does not identify the Koblitz degrees: the recorded factorization 703889=409·1721 and volcano hypotheses are additionally required.

Evidence impact

The new diagram is an audit workflow, not a quantitative result. Existing historical figure values, parent-study report/PDF and benchmark mirrors are preserved. A corrected native reanalysis must refresh all affected canonical graphs and copies together. No new timing, speedup or leaderboard row is introduced. Full findings, exact pins and validation receipts are in REVIEW-20261019.md and PROOF-MAP.md beside the manuscript.